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AIVA: UNIFIED AI INTERACTION SUITE

Description

This project is an advanced, multi-modal system designed as the Artificially Intelligent Virtual Assistant (AIVA), unifying several AI and computer vision technologies into one intelligent control hub. AIVA eliminates the reliance on traditional input devices by enabling users to interact with the computer using voice commands, hand gestures, head movements, and a virtual keyboard. It serves as a centralized solution to enhance accessibility for users with mobility impairments and boost efficiency for all users by offering a seamless, hands-free computing experience. The core objective is to create a natural and adaptive mode of human-computer interaction, laying the foundation for future touchless systems.

KEY MODULES

1. Proton (Voice Assistant): Utilizes the Google Gemini API, speech recognition, and Text-to-Speech (TTS) to understand and execute complex verbal commands, automating system tasks and providing intelligent responses.

2. Gesture Mouse: Employs real-time hand-tracking via computer vision to translate gestures (e.g., pinching, finger movements) into precise mouse control, clicks, and drag operations.

3. Head Tracker: Maps facial landmarks and head movements to the cursor position, offering an essential accessibility feature for hands-free navigation.

4. Virtual Keyboard: A contactless, dwell-time-based on-screen keyboard for secure and accessible text entry without physical hardware.

TECHNOLOGIES USED

- **Core Language:** Python (Used for system stability, real-time processing, and leveraging the extensive AI library ecosystem).
- **Computer Vision & Tracking:** OpenCV and MediaPipe (For highly accurate hand and face landmark detection).
- **AI Integration:** Google Gemini API (For sophisticated Natural Language Processing and conversational intelligence).
- **Speech & Audio:** SpeechRecognition, pyttsx3, and Pygame (For voice I/O and multimedia control).
- **GUI & Input Control:** Tkinter and pynput (For the main Control Hub dashboard and low-level keyboard/mouse simulation).